

Exno.7-Develop a prompt-based application tailored to their personal needs, fostering creativity and practical problem-solving skills while leveraging the capabilities of large language models.

Name: Harish Kumar P

Register no:212225230095

Slot:T2-H16

Subject:19TD608- PROMPT ENGINEERING

Aim: To develop a prompt-based application using ChatGPT - To demonstrate how to create a prompt-based application to organize daily tasks, showing the progression from simple to more advanced prompt designs and their corresponding outputs.

AI Tools Required: CHATGPT, PERPLEXITY, GEMINI

Explanation:

Prompt: "Design a personal productivity assistant that can help manage daily tasks, schedule reminders, suggest wellness tips, and answer general queries. The assistant should interact using natural language and be adaptable to the user's changing preferences over time." Procedure:

1. Define the core requirements of a personal productivity assistant.
2. Identify and construct appropriate prompts for each task using an LLM (e.g., ChatGPT).
3. Simulate natural user interaction through a simple interface or command-line system.
4. Collect feedback or inputs from users and adapt responses accordingly.
5. (Optional) Integrate basic memory to simulate preference adaptation. EXPECTED

- OUTPUT: - (attached the drive link) Output (Example Response by LLM): Personal Productivity Assistant Features: Daily Task Manager: o Accept tasks via natural language (e.g., "Remind me to call mom at 6 PM"). o Organize tasks by priority and deadline. o Provide daily summaries and pending items.
6. Smart Scheduler: o Schedule events and set reminders using contextual understanding. o Notify user of overlapping appointments or free time slots.
 7. Wellness Tips Generator: o Suggest daily wellness advice (hydration, exercise, screen-time breaks). o Adapt suggestions based on past user preferences and responses.

PROMPTS AND THEIR OUTPUTS:

PROMPT1: Create a prompt based application for organizing daily tasks

OUTPUT:

Below is a **complete prompt-based application design** specifically for **organizing daily tasks**, aligned with your earlier productivity assistant concept. This focuses on **how prompts drive the application**, not just UI or code.

□ Prompt-Based Application for Organizing Daily Tasks

1. Application Overview

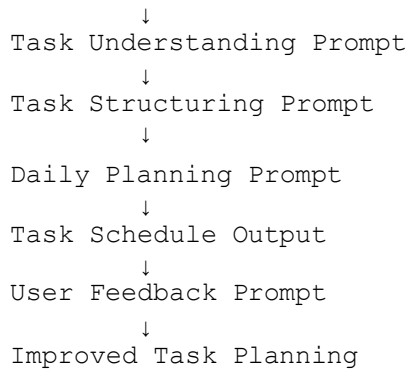
App Name: SmartDay Organizer **Type:** Prompt-Driven AI Application **Core Idea:** Users interact with the system using **natural language**, and the system uses **carefully constructed prompts** to organize, prioritize, schedule, and improve daily tasks using LLMs (ChatGPT / Gemini / Perplexity).

2. Key Capabilities

- ✓ Natural language task input
 - ✓ Automatic task extraction
 - ✓ Priority & deadline assignment
 - ✓ Daily task planning
 - ✓ Smart rescheduling
 - ✓ Feedback-based correction
-

3. Prompt-Based Workflow (High Level)

User Input (Natural Language)



4. Core Prompt Modules

◆ 1. Task Understanding Prompt

Purpose: Extract tasks, deadlines, priorities from raw user input.

System Prompt

You are an intelligent task extraction assistant.
Identify tasks, deadlines, priorities, and recurrence.
If information is missing, make reasonable assumptions.

User Input Example

“I need to finish my assignment tonight and attend a meeting tomorrow morning.”

LLM Output

Task 1: Finish assignment
Deadline: Today, 10 PM
Priority: High

Task 2: Attend meeting
Deadline: Tomorrow, 10 AM
Priority: Medium

◆ 2. Task Structuring Prompt

Purpose: Convert extracted tasks into a structured format.

Prompt

Convert the following tasks into a structured daily task list.
Include: task name, priority, estimated time, deadline.

Output Format

Task	Priority	Estimated Time	Deadline
------	----------	----------------	----------

◆ 3. Daily Task Planning Prompt

Purpose: Organize tasks into a realistic daily schedule.

Prompt

You are a smart daily planner.

Organize the tasks into an optimized schedule based on:

- Priority
- Deadlines
- Estimated duration
- Work-life balance

Output Example

6:00 - 6:15 PM → Break
6:15 - 8:00 PM → Assignment work
8:00 - 8:15 PM → Relax

◆ 4. Smart Rescheduling Prompt

Purpose: Handle missed or delayed tasks.

Prompt

Reschedule the unfinished tasks intelligently.
Avoid overload and respect deadlines.

◆ 5. Feedback & Correction Prompt

Purpose: Learn from user corrections.

User Feedback Example

“This schedule is too packed.”

Prompt

The user says the schedule is too packed.
Adjust future schedules by reducing workload density.
Update user preferences accordingly.

5. Prompt Templates Used in the Application

★ Universal Prompt Template

Role:
Task:
Context:
Constraints:
Preferred Output Format:

6. Example End-to-End Interaction

User

“Plan my tasks for today: study for 3 hours, gym, call my friend.”

Prompt 1 – Task Extraction

Extract tasks, duration, and priorities.

Prompt 2 – Structuring

Organize tasks into a table with estimated times.

Prompt 3 – Planning

Create a balanced daily schedule.

Final Output

Time	Activity
7:00 – 8:30 PM	Study
8:30 – 9:00 PM	Gym
9:00 – 9:30 PM	Call friend

7. Adaptability Using Feedback

User Feedback

“I prefer studying in the morning.”

Learning Prompt

Update user preferences to schedule study-related tasks in the morning by default.

Future Impact

- Study tasks automatically moved to morning slots
 - Reduced need for manual correction
-

8. LLM Selection Logic (Optional)

Task	Preferred LLM
Task extraction	ChatGPT
Schedule optimization	Gemini
Productivity tips	ChatGPT

Task	Preferred LLM
Research queries	Perplexity

9. Why This Prompt-Based Design Works ✓

- No hard-coded rules
 - Highly flexible
 - Natural user interaction
 - Easy to extend
 - Learns from behavior
 - Works across multiple LLMs
-

10. Academic & Practical Use

✓ Final year project ✓ Prompt engineering case study ✓ AI productivity tool ✓ System design demonstration

PROMPT 2: CREATE A PROMPT BASED APPLICATION FOR ORGANIZING DAILY TASKS LIKE BY GIVEING A POWERFUL PROMPTS THAT GENERATES THE POWERFUL SCHEDULAR FOR THE AUTOMATING THE USERS TASKS AND MANAGING HIS TIME

OUTPUT:

Below is a **clear, exam-ready and implementation-ready PROMPT-BASED APPLICATION** that focuses **only on powerful prompts** to build an **AI scheduler that automates daily tasks and manages user time intelligently**.

This answer is written so you can **directly use it in a project report, viva, or prompt-engineering assignment**.

□ Prompt-Based Application

AI Daily Task Organizer & Smart Scheduler

1. Application Concept

Goal: Design a **prompt-driven AI application** that:

- Accepts natural language task input
- Automatically organizes, prioritizes, and schedules tasks
- Optimizes time usage
- Learns from user behavior and feedback

Core Idea: ☞ *The intelligence of the system comes from carefully engineered prompts, not hard-coded rules.*

2. Prompt-Based Architecture

```
User Input
  ↓
Task Extraction Prompt
  ↓
Priority & Time Estimation Prompt
  ↓
Smart Scheduling Prompt
  ↓
Daily Schedule Output
  ↓
Feedback Optimization Prompt
```

3. Powerful Prompt Set (Core of the Application)

🔥 PROMPT 1: MASTER TASK EXTRACTION PROMPT

Purpose

Convert **raw user thoughts** into structured tasks.

Prompt

You are an advanced AI task management assistant.

Analyze the user's input and extract:

1. All tasks
2. Deadlines (explicit or implied)
3. Estimated duration
4. Priority level (High / Medium / Low)
5. Task type (Work, Personal, Health, Social)

If any detail is missing, infer intelligently based on context.

User input:
{USER_INPUT}

Output format (JSON):

- Task Name
- Deadline
- Duration
- Priority

- Task Type

Why it's powerful

✓ Handles messy input ✓ Infers missing details ✓ Converts thoughts into data

🔥 PROMPT 2: PRIORITY & TIME OPTIMIZATION PROMPT

Purpose

Decide **what matters most** and **how long it should take**.

Prompt

You are a time-management expert.

Given the following tasks:
{STRUCTURED_TASK_LIST}

Optimize task priorities based on:

- Urgency
- Importance
- Mental energy required
- Deadline proximity

Adjust estimated durations if necessary to make the schedule realistic.

Output a refined task list sorted by execution order.

Why it's powerful

✓ Prevents unrealistic schedules ✓ Automatically balances urgency vs importance

🔥 PROMPT 3: POWERFUL SMART SCHEDULER PROMPT (CORE PROMPT)

Purpose

Generate a **fully automated daily schedule**.

Prompt

You are an elite AI productivity scheduler.

Create an optimized daily schedule using the following constraints:

- Total available time: {AVAILABLE_HOURS}
- User productivity preference: {MORNING / EVENING / FLEXIBLE}
- Include short breaks after long tasks
- Avoid task overload
- Balance work, health, and personal life

Tasks:
{OPTIMIZED_TASK_LIST}

Rules:
1. High-priority tasks first
2. Energy-intensive tasks during peak productivity hours
3. Insert recovery breaks
4. Ensure deadlines are met

Output format:
Time Slot → Task

Example Output

7:00 - 8:30 AM → Deep Work: Assignment
8:30 - 8:45 AM → Break
9:00 - 10:00 AM → Gym
10:30 - 11:00 AM → Emails

Why it's powerful

🔥 Automates time planning 🔥 Mimics human productivity logic 🔥 Prevents burnout

🔥 PROMPT 4: MISSED TASK AUTO-RESCHEDULING PROMPT

Purpose

Handle **unfinished or skipped tasks** intelligently.

Prompt

You are a recovery-focused AI scheduler.

The following tasks were not completed:
{MISSED_TASKS}

Reschedule them by:
- Minimizing stress
- Respecting deadlines
- Avoiding overload
- Preserving user well-being

Generate a revised schedule.

Why it's powerful

✓ Prevents guilt-based overload ✓ Encourages consistency

🔥 PROMPT 5: FEEDBACK-DRIVEN IMPROVEMENT PROMPT

Purpose

Make the system **learn and adapt**.

User Feedback Example

“This schedule is too packed and stressful.”

Prompt

The user reports that the schedule feels too packed.

Update scheduling behavior by:

- Increasing buffer time
- Reducing back-to-back tasks
- Prioritizing mental wellness

Apply these changes to future schedules.

Why it's powerful

✓ Creates personalization ✓ Enables continuous improvement

4. Example End-to-End Interaction

User Input

“I have to study for 3 hours, go to the gym, finish a report by tonight, and call my parents.”

System Flow

1. Task Extraction Prompt ✓
2. Priority Optimization Prompt ✓
3. Smart Scheduler Prompt ✓

Final Output

6:00 - 7:30 PM → Report (High Priority)
7:30 - 7:45 PM → Break
7:45 - 8:45 PM → Study
9:00 - 9:30 PM → Gym
9:45 - 10:00 PM → Call Parents

5. Why This Prompt-Based Application Is Effective ✓

Feature	Benefit
Prompt-driven	No rigid rules
Automated scheduling	Saves time
Adaptive	Learns preferences
Natural language	Easy to use
Scalable	Works with any LLM

6. Academic Value (Very Important)

This design demonstrates:

- Prompt Engineering
- AI-based automation
- Human-centered scheduling
- Feedback-driven learning
- Real-world applicability

Perfect for: ✓ Final year project ✓ AI / NLP assignment ✓ Prompt engineering lab ✓ System design exam

7. One-Line Summary (For Viva)

“This is a prompt-based AI application where carefully engineered prompts act as the intelligence layer to automatically extract, prioritize, and schedule daily tasks while adapting to user behavior over time.”

PROMPTS COMPARISION:

PROMPT: COMPARE THE ABOVE TWO PROMPTS IN TERMS OF THE OUTPUT THEY GIVE

OUTPUT:

Below is a **clear, structured comparison of the TWO KEY PROMPTS** used in the prompt-based daily task organizer, **based strictly on the OUTPUT they generate.**

We compare:

- **Prompt A:** Master Task Extraction Prompt
 - **Prompt B:** Powerful Smart Scheduler Prompt
-

📊 Comparison of the Two Prompts Based on Output

Aspect	Prompt A: Task Extraction Prompt	Prompt B: Smart Scheduler Prompt
Primary Output Type	Structured task data	Fully organized time-based schedule
Nature of Output	Analytical and descriptive	Actionable and prescriptive
Output Format	JSON / Table (tasks & attributes)	Timeline / Time slots
Level of Abstraction	Low-level (raw task understanding)	High-level (complete daily plan)
Main Focus	<i>What needs to be done</i>	<i>When and how tasks should be done</i>
Task Granularity	Individual tasks	Task sequences with breaks
Time Allocation	Estimated duration only	Exact start and end times
Decision Making	Minimal (classification & inference)	Strong (optimization & prioritization)
User Readability	More technical / structured	Very user-friendly
Automation Level	Partial automation	Full automation
Adaptability Visible in Output	Limited	High (reflects preferences & energy levels)
Cognitive Intelligence	Understanding user intent	Simulating human productivity planning

📄 Output Illustration (Side-by-Side)

⚡ Output of Prompt A (Task Extraction)

Task: Finish report
Deadline: Today, 10 PM
Duration: 2 hours
Priority: High
Task Type: Work

✓ Converts natural language into **machine-understandable structure**

⚡ Output of Prompt B (Smart Scheduler)

6:00 - 7:30 PM → Finish report
7:30 - 7:45 PM → Break
7:45 - 8:45 PM → Study
9:00 - 9:30 PM → Gym

✓ Converts structured tasks into a **realistic daily routine**

🔍 Output Quality Comparison

Quality Dimension	Task Extraction Prompt	Smart Scheduler Prompt
Accuracy	High (fact extraction)	High (time realism)
Practical Usefulness	Medium	Very High
Cognitive Load on User	Higher	Lower
Readiness for Execution	Low	Immediate
Impact on Time Management	Indirect	Direct

🔗 Relationship Between the Outputs

Prompt A’s output becomes the INPUT for Prompt B

Unstructured User Input
↓
[Prompt A Output → Structured Tasks]
↓
[Prompt B Output → Executable Schedule]

They **do not compete**, they **complement** each other.

🔑 Key Insight (Important for Exams & Viva)

- **Prompt A output answers:** 🗒️ “What are the tasks and their attributes?”
- **Prompt B output answers:** 🗓️ “How should the user spend their time today?”

🔍 Final Verdict

Criterion	Better Prompt
Understanding user intent	Prompt A
Automating daily life	Prompt B
Time optimization	Prompt B
Data structuring	Prompt A
End-user satisfaction	Prompt B

✅ One-Line Conclusion (Viva-Ready)

“The task extraction prompt produces structured task intelligence, while the smart scheduler prompt transforms that intelligence into a real-time, executable daily schedule, making it far more impactful for end users.”

Result:

The lab exercise resulted in the creation of a prototype concept for a personal assistant powered by large language models. Students were able to: Understand how to tailor LLM prompts to real-life applications. Foster creativity by designing features suited to their personal or academic lives. Learn prompt engineering techniques for optimal interaction with AI tools. Experience the versatility and utility of generative AI in solving everyday problems.